

Seismograms

Seismograms record the arrival of seismic rays at a location distant from an earthquake

The travel-time (t) of a seismic ray is proportional to the length (x) of the travel path and velocity (v) of the rocks it travels through,

$$t \propto x v^{-1}$$

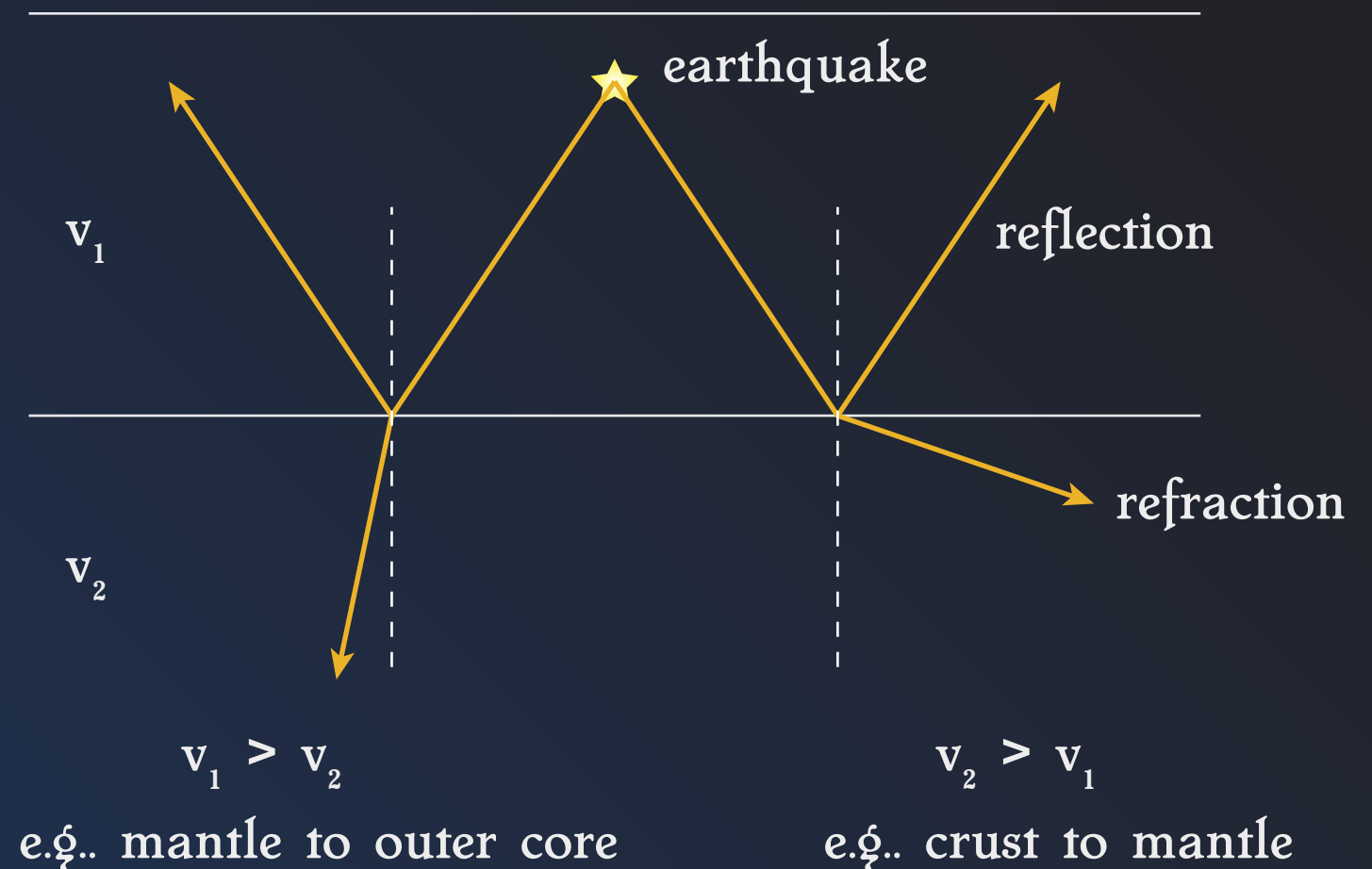
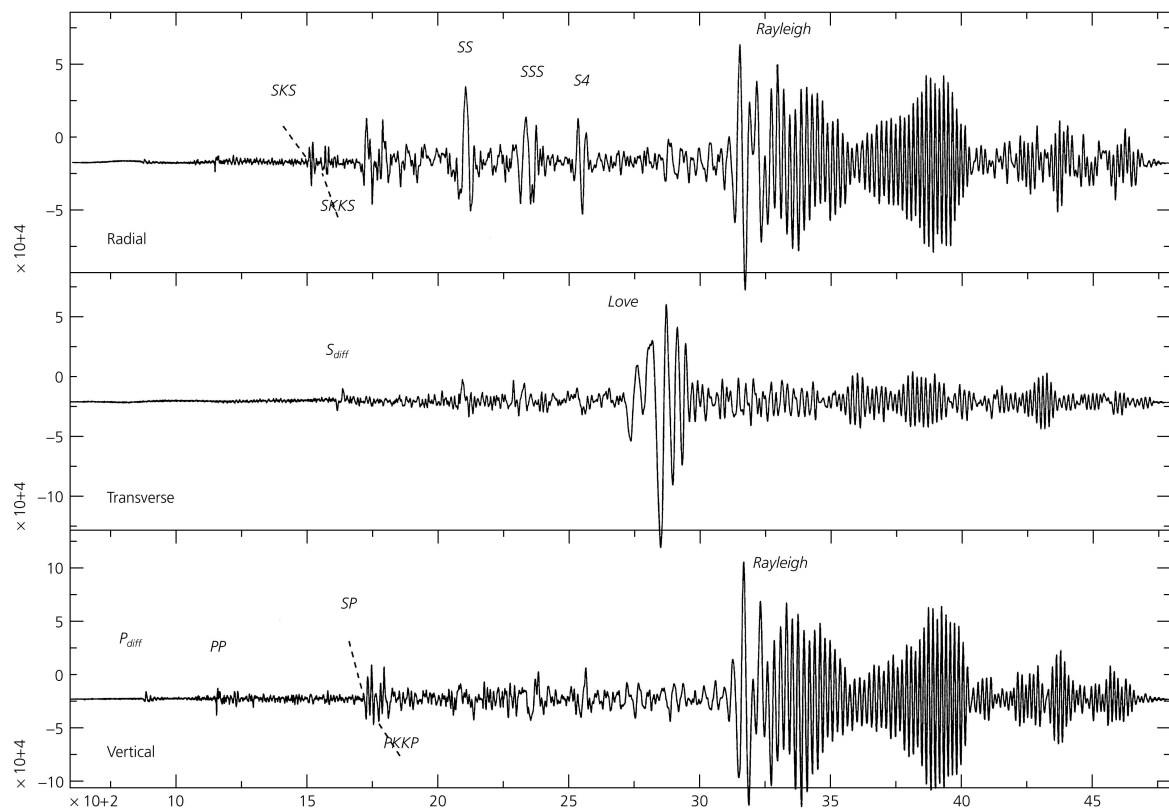


Figure 2.7-1: Seismograms recorded at a distance of 110°, showing surface waves.



Seismic rays are bent (refracted) like light passing through a prism or reflected like light bouncing off a mirror

Refracted waves bend in a predictable way according to Snell's law